

Intelligent Medical Report Analysis System

Project Domain: DL / GenAI / NLP
DSAI LAB Group 2



Indian Institute Of Technology, Madras

TEAM

01

Shivendra Patel
21f2001310@ds.study.iitm.ac.in
S.P.

02

Samta Ranka
23f1001316@ds.study.iitm.ac.in
S.R.

03

Rajat Srivastava
22f3003195@ds.study.iitm.ac.in
R.S

04

Ritwik Trivedi
22f1000120@ds.study.iitm.ac.in
R.T.W

05

Bryan David Robinson
22f3002277@ds.study.iitm.ac.in
B.D.R

Introduction-Dataset Description & Preprocessing

Datasets

- **MIMIC-III:** Clinical EHR and laboratory records for real-world medical data.
- **MTSamples:** Medical transcription reports from multiple specialties for diverse report formats.

Preprocessing

- **OCR Extraction (PaddleOCR):** Extracted text from scanned lab reports.
- **Table-to-Narrative Conversion:** Standardized tabular data into text (e.g., *WBC 4.0 ×10⁹/L*).
- **BIO Annotation:** Token-level labeling using **O, B/I-TEST, B-VALUE, B/I-UNIT, B-REFERENCE**.

Output: Clean, standardized, BIO-annotated dataset ready for ClinicalBERT training.

Medical Report Image → PaddleOCR → ClinicalBERT (NER) → Structured JSON → BioMistral 7B → Patient-Friendly Summary

Model Architecture Overview

- 1.PaddleOCR:** Extracts text from medical report images.
- 2.ClinicalBERT (NER):** Identifies laboratory entities using BIO tagging (Test, Value, Unit, Reference Range).
- 3.Structured Data Aggregation:** Organizes extracted entities into a standardized JSON format.
- 4.BioMistral 7B:** Generates patient-friendly summaries and explains laboratory results from the structured data.

Information Extraction Engine (ClinicalBERT Token Classifier)

- **Base Model:** ClinicalBERT (medicalai/ClinicalBERT)
- **Objective / Loss:** Cross-Entropy Loss with padding masking (ignore_index = -100)
- **Target Tags:** ['O', 'B-TEST', 'I-TEST', 'B-VALUE', 'B-UNIT', 'I-UNIT', 'B-REFERENCE']
- **Metrics:** Token-level and Entity-level Precision, Recall, and Strict F1-score (evaluated via seqeval)
- **Optimizer:** AdamW (LR = 0.00002, Weight Decay = 0.01)
- **Scheduler:** Linear Warmup over 10% of total steps
- **Hardware:** CUDA-enabled GPU (Tesla T4/P100 environment)

Patient-Friendly LLM Summarizer (BioMistral 7B)

- **Base Model:** BioMistral 7B
- **Fine-tuning Technique:** *QLoRA 4-bit*
- **Loss Function:** *Negative Log Likelihood*
- **Optimizer & Schedule:** *AdamW_torch_fused , Cosine Annealing*
- **Epochs / Steps & Batch Size:** *3 Epochs, train batch size = 2, Gradient Accumulation = 4*
- **Hardware Used:** *2 * T4 GPUs*

Training Setup (ClinicalBERT)

- 1.Learning Rate:** 2×10^{-5}
- Optimizer:** AdamW with **0.01** weight decay
- 2.Training:** 10 epochs with a batch size of **32** for both training and evaluation
- 3.Evaluation & Checkpointing:** Performed after every epoch, with the best model selected based on the lowest validation loss
- 4.Learning Rate Scheduling:** Linear warmup over **10%** of the total training steps
- 5.Mixed Precision:** FP16 enabled for faster training and reduced GPU memory usage
- 6.Logging:** Training progress recorded every **100** steps
- 7.Checkpoint Management:** Maximum of **3** model checkpoints retained during training

Resource	Specification
GPU	NVIDIA Tesla T4
Memory	16 GB GPU RAM
CPU	4 cores
Training Time	~4-6 hours
Framework	PyTorch / Hugging Face transformers

Fine-Tuning & Experiments

ClinicalBERT Fine-Tuning

- Fine-tuned **ClinicalBERT** for token-level laboratory entity extraction using the **BIO tagging scheme**.
- Trained on **15138 reports** using **AdamW**, **10 epochs**, **batch size 32**, and **learning rate 2×10^{-5}** .
- Applied **weighted cross-entropy loss** with inverse square-root class weighting to address class imbalance.
- Achieved **99% test accuracy** and **99% Entity F1-score**, demonstrating highly accurate extraction of lab tests, values, and units.

Metric	Calculation	Purpose
Accuracy	$(TP + TN) / \text{Total}$	Overall performance
F1-Score	$2 * (P * R) / (P + R)$	Entity-level performance
Precision	$TP / (TP + FP)$	Minimize false positives
Recall	$TP / (TP + FN)$	Minimize false negatives
Macro F1	Average per entity	Balanced performance
Weighted F1	Weighted by support	Class imbalance handling

BioMistral-7B Fine-Tuning

- Fine-tuned **BioMistral-7B** to generate patient-friendly explanations from **ClinicalBERT's extracted laboratory information**.
- Built an instruction-tuning dataset of **92 MIMIC-III examples** using **knowledge distillation** from NVIDIA Nemotron-3-Ultra-550B.
- Used **4-bit NF4 QLoRA** with LoRA adapters, training only **20.97M (0.2888%)** of the **7.26B** model parameters.
- Performed **Optuna-based hyperparameter optimization**, selecting the best configuration (**r = 8**, **α = 16**, **dropout = 0.1879**, **learning rate = 3.31×10^{-4}**) with the **lowest evaluation loss (0.6906)**.

ClinicalBert Performance

- Overall Token Accuracy: 99%
- Entity-Level Macro F1-Score: 99%
- Overall Weighted F1-Score: 99%
- Test Loss: 0.00239

Qualitative Evaluation

BioMistral 7BQunatitavie Evaluation

Metric	Selected model (Trial 5)	Notes
Eval loss (NLL)	0.6906	Lowest of the 8 trials
Perplexity	1.99	exp(0.6906)
Eval mean token accuracy	0.8075	Next-token agreement with reference
Eval entropy	0.5387	Lower indicates more confident predictions
Train loss	0.6319	Train-to-eval gap of 0.06 (good generalisation)
Trainable parameters	20,971,520 (0.2888%)	LoRA adapter only; base weights frozen
Evaluation set size	19 examples	Small; treat figures as indicative

Observation	Details
Multi-word tests	Correctly parses complex clinical terms like "White Blood Cells" and "Alkaline Phosphatase".
Units	Near-perfect detection across diverse units (g/dL, 10^9/L, mEq/L, %).
Values	Excellent boundary detection for floating point and integer test results.
Reference ranges	Lower recall on reference ranges due to formatting variations across different hospital labs.
Context understanding	Successfully resolves shorthand variations (e.g., Hgb vs. Hemoglobin, WBC vs. White Blood Count).
OCR Mapping	Effectively aligns raw OCR outputs into clean JSON key-value pairs.

Regularization & Training Stability

Technique	Description	Impact
Optimizer (AdamW)	Weight decay-decoupled adaptive optimizer ($\text{lr}=2\text{e-}5$, $\text{weight_decay}=0.01$)	Prevents overfitting
Weight Decay	L2 regularization (0.01)	Improves generalization
Warmup Steps	Linear warmup for first 10% of steps	Stabilizes early training
Gradient Clipping	Max norm = 1.0	Prevents gradient explosion
Mixed Precision (FP16)	16-bit floating point	Faster training, less memory
Dropout	Randomly zeroes activations ($p=0.1$) in attention and hidden layers	Reduces overfitting
Best Model Selection	Restores weights from the epoch with the lowest validation loss	Ensures best validation performance

Sample Output from End-to-End Pipeline

Structured Output

Sample Input Text

hemoglobin 15.2 g/dL
hct 44.1 %
platelets 280 10⁹/L
mcv 89 fL
neutrophils 62 %
lymphocytes 28 %
glucose 98 mg/dL
sodium 141 mmol/L
potassium 4.3 mmol/L
chloride 105 mmol/L
bicarbonate 28 mmol/L
bun 16 mg/dL
creatinine 1.05 mg/dL
calcium 9.7 mg/dL
cholesterol 215 mg/dL
hdl 42 mg/dL
ldl 145 mg/dL
triglycerides 138 mg/dL

Token-Level Predictions

LABORATORY -> O
RESULTS -> O
-----> O
wbc -> O
40 -> B-VALUE
10⁹/L -> O
hemoglobin -> B-TEST
15.2 -> B-VALUE
g/dL -> B-UNIT
hct -> B-TEST
44.1 -> B-VALUE
% -> O
platelets -> B-TEST
280 -> B-VALUE
10⁹/L -> B-VALUE
mcv -> B-TEST
89 -> B-VALUE
fL -> B-UNIT
neutrophils -> B-TEST
62 -> B-VALUE
% -> O
lymphocytes -> B-TEST

```
{  "test": "hemoglobin",  
    "value": "15.2",  
    "unit": "g/dL",  
    "test_name": "Hemoglobin",  
    "status": "NORMAL"  
}, {  
    "test": "hct",  
    "value": "44.1",  
    "unit": "%",  
    "test_name": "Hematocrit",  
    "status": "NORMAL"}
```

St. Jude's Diagnostic Laboratories		LABORATORY TEST RESULTS		
Lab Test Results Report		REPORT DATE: 2024-05-15		
PATIENT INFORMATION				
PATIENT NAME: DOE, JOHN A.		SEX: M		
DATE OF BIRTH: 1978-08-21 (AGE: 45)		PATIENT ID: PT-339-8810		
REFERRING PHYSICIAN: DR. ELIZABETH CHEN				
TEST DETAILS				
ORDER DATE: 2024-05-14		SPECIMEN TYPE: BLOOD, VENOUS		
TEST RESULTS				
Test Name	Result	Units	Reference Range	Flags
Complete Blood Count (CBC)				
WBC Count	8.1	10 ⁹ /L	4.0 - 11.0	
Hemoglobin	15.2	g/dL	13.5 - 17.5	
Hematocrit	44.1	%	41.0 - 53.0	
Platelet Count	280	10 ⁹ /L	150 - 450	
MCV	89	fL	80 - 100	
Neutrophils	62	%	40 - 75	
Lymphocytes	28	%	20 - 45	
Basic Metabolic Panel (BMP)				
Glucose, Fasting	88	mg/dL	70 - 99	
Sodium (Na ⁺)	141	mmol/L	135 - 145	
Potassium (K ⁺)	4.3	mmol/L	3.5 - 5.1	
Chloride (Cl ⁻)	105	mmol/L	98 - 107	
Bicarbonate (CO ₃)	28	mmol/L	22 - 29	
BUN (Blood Urea Nitrogen)	16	mg/dL	7 - 20	
Creatinine	1.05	mg/dL	0.70 - 1.30	
Calcium	9.7	mg/dL	8.5 - 10.5	
Lipid Panel				
Total Cholesterol	215	mg/dL	< 200	HIGH
HDL Cholesterol	42	mg/dL	> 40	
LDL Cholesterol (Calc.)	145	mg/dL	< 130	HIGH
Triglycerides	138	mg/dL	< 150	
COMMENTS / INTERPRETATION				
Total cholesterol and LDL cholesterol are moderately elevated. Recommend correlation with clinical history and discussion regarding diet/lifestyle modifications.				
LAB SUPERVISOR: DR. DAVID PATEL, PH.D., HCLD ELECTRONICALLY SIGNED AND RELEASED: 2024-05-15 14:30 EST Page 1 of 1				

THANKYOU

